

# A260 – The Casimir Effect in FFGFT: Vacuum Structure and Experimental Tests

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## .1 Purpose

**[STIPULATION]** This document treats the Casimir effect within the FFGFT framework. The Casimir effect is in the legacy corpus the only experimentally accessible bridge between the fundamental length scale  $L_0 = \xi \cdot \ell_P$  and measurable vacuum forces. The document shows: (1) The modified Casimir formula reproduces the standard formula exactly – no new physics at currently accessible separations. (2) The prefactor 4/3 in  $\xi = 4/3 \times 10^{-4}$  is confirmed by the Casimir effect independently of mass fits (A015). (3) There is a characteristic vacuum length scale  $L_\xi \approx 100 \mu\text{m}$  at which Casimir energy density and a specific vacuum quantity coincide – this is here explicitly excluded.

## .2 Fundamental Length Scales

**[B]** FFGFT defines a hierarchy of characteristic length scales:

$$\begin{aligned}L_0 &= \xi \cdot \ell_P \approx 2.155 \times 10^{-39} \text{ m}, \\ \ell_P &= \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35} \text{ m}, \\ L_\xi &\approx 100 \mu\text{m}.\end{aligned}$$

Sub-Planck (spacetime granulation), Planck length, and characteristic vacuum length scale (measurable). The coupling constant  $\xi = 4/3 \times 10^{-4}$  is the same parameter as in A020. It can be derived from a Lagrangian describing the dynamics of the time field (A142).

**[B] Spacetime granulation.** At  $L_0$  all vacuum fluctuations are fully active. For  $d > L_0$  only parts of these forces are visible – this is the physical origin of the  $1/d^4$  dependence of the Casimir force. Due to the extremely small size of  $L_0 \approx 2 \times 10^{-39} \text{ m}$ , direct measurement is not possible.

### .3 The Modified Casimir Formula and Its Consistency

**[K]** The modified Casimir energy density in the FFGFT framework:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{720 \xi} \rho_{\text{vac}} \cdot \left(\frac{L_\xi}{d}\right)^4,$$

where  $\rho_{\text{vac}} = \xi \hbar c / L_\xi^4$  is a characteristic vacuum density. Substituting:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{720 \xi} \cdot \frac{\xi \hbar c}{L_\xi^4} \cdot \frac{L_\xi^4}{d^4} = \frac{\pi^2 \hbar c}{720 d^4}.$$

(Corrected on 1 Oct. 2026: previously 240 instead of 720 in both formulas –  $\pi^2 \hbar c / (240 d^4)$  is the Casimir pressure; the energy density between the plates is  $\pi^2 \hbar c / (720 d^4)$ , as corrected in Doc. 061.) This is exactly the standard Casimir formula (energy density). No contradiction, no new parameter – the FFGFT formulation is numerically equivalent to the established QED results.

**[K] Numerical verification** (Casimir energy density vs. plate separation):

| Separation $d$    | $\rho_{\text{Casimir}}$ (J/m <sup>3</sup> ) | Ratio to $\rho_{\text{vac}}$ |
|-------------------|---------------------------------------------|------------------------------|
| 100 $\mu\text{m}$ | $4.33 \times 10^{-12}$                      | 102.8                        |
| 10 $\mu\text{m}$  | $4.33 \times 10^{-8}$                       | $1.03 \times 10^6$           |
| 1 $\mu\text{m}$   | $4.33 \times 10^{-4}$                       | $1.03 \times 10^{10}$        |

(Corrected on 1 Oct. 2026: previously  $4.17 \times 10^{-14}$  /  $4.17 \times 10^{-10}$  /  $4.17 \times 10^{-2}$  J/m<sup>3</sup> and ratios 1.00 /  $10^4$  /  $10^{12}$  – the 100- $\mu\text{m}$  row gave  $\rho_{\text{vac}}$  instead of  $\rho_{\text{Casimir}}$ , and from 10 to 1  $\mu\text{m}$  a factor  $10^8$  instead of  $10^4$  ( $d^{-4}$ ) had been used. Recomputed:  $\pi^2 \hbar c / (720 d^4) = 4.33 \times 10^{-12}$ ,  $4.33 \times 10^{-8}$ ,  $4.33 \times 10^{-4}$  J/m<sup>3</sup>; with  $\rho_{\text{vac}} = \xi \hbar c / L_\xi^4 = 4.22 \times 10^{-14}$  J/m<sup>3</sup> ( $L_\xi = 10^{-4}$  m) the ratio is  $\pi^2 / (720 \xi) (L_\xi / d)^4 = 102.8$ ,  $1.03 \times 10^6$ ,  $1.03 \times 10^{10}$ .)

**[K] Characteristic scale hierarchy:**

$$\frac{L_0}{\ell_P} = \xi = 1.333 \times 10^{-4}, \quad \frac{\ell_P}{L_\xi} \approx 1.6 \times 10^{-31}.$$

### .4 Confirmation of the Factor 4/3

**[B]** The prefactor 4/3 in  $\xi = 4/3 \times 10^{-4}$  appears independently in the Casimir context: the factor 4/3 appears in the Casimir energy term as  $E_{\text{Casimir}} \propto \frac{4}{3}$  (the fourth – musical interval 4:3) and is therefore not only a mass-fit quantity. This independence is recorded in A015 as Casimir confirmation: no other prefactor (fifth 3/2, third 5/4) gives consistent results for the mass spectrum and the Casimir structure simultaneously.

### .5 Experimental Tests

**[K]** Three classes of tests:

**Precision measurements at  $d \approx L_\xi$ :** At plate separation  $\approx 100 \mu\text{m}$  the Casimir energy density reaches the order of  $\rho_{\text{vac}}$ . This regime is experimentally accessible. (Note of 1 Oct. 2026: with the energy density  $\pi^2 \hbar c / (720 d^4)$ ,  $\rho_{\text{Casimir}}$  at  $d = 100 \mu\text{m}$  exceeds  $\rho_{\text{vac}}$  by a factor of 102.8; equality  $\rho_{\text{Casimir}} = \rho_{\text{vac}}$  is reached only at  $d = L_\xi (\pi^2 / (720 \xi))^{1/4} \approx 318 \mu\text{m}$ .)

**Scaling behaviour:** The  $1/d^4$  dependence should be exactly satisfied down into the nanometre range. Deviations at  $d \lesssim 10$  nm could indicate spacetime granulation.

**Experimental  $L_\xi$  values from the literature:** Parallel plates:  $\sim 228$  nm; sphere-plate geometry:  $\sim 1.75 \mu\text{m}$  to  $\sim 18 \mu\text{m}$ . The scatter reflects geometric differences ( $F \propto 1/d^4$  vs.  $F \propto 1/d^3$ ).

## .6 Verification Script

- Numerical verification of the modified Casimir formula (reproduction of the standard formula), length scale hierarchy, comparison with  $L_\xi$  values: `python/A_Serie_Skripte/a260_casimir.py`

## .7 Open Questions

**Direct measurement of  $L_0$  is not possible.** The minimal length scale  $L_0 \approx 2 \times 10^{-39}$  m lies far beyond any experimental reach. The characteristic scale  $L_\xi$  is experimentally accessible, but its precise connection to  $L_0$  presupposes the FFGFT framework.

**Deviations from the  $1/d^4$  law at  $d \sim 10$  nm are not measured.** The prediction of a scale change near the sub-Planck boundary is qualitative, not quantitatively worked out.

## .8 Supersedes

This document subsumes: Doc. 091 (modified Casimir theory, length scale hierarchy, numerical verification, experimental predictions and tests). *Not* taken over: the Casimir vacuum scaling from Doc. 091 (excluded sector, outside A-series core) and the macroscopic vacuum implications.

## .9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.